Consensus U-Nets, a Seed Refiner and a Learned Filter for Solar Filament Segmentation

A Solution to the Solar Filament Segmentation Challenge 2026

Nhat Vo
Sensen.ai
Melbourne, Australia
mr.nhatvo@gmail.com

ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. The training labels are noisy: annotators of the same image agree with each other at only PQ 0.34. Ten U-Nets (five folds at 1024 and five at 2048 pixels) learn the per-pixel mean of all annotators' masks; a simple post-processing step turns their mean map into candidate filaments; a seed refiner re-segments a crop around each candidate; and a learned filter removes candidates that annotators are unlikely to mark. Out of fold, the pipeline reaches PQ 0.445 on all 707 training images (95% bootstrap interval 0.435–0.454), against 0.413 for a single U-Net. The public leaderboard score is 0.38. The code, the weights and a notebook that rebuilds the submission are in our repository.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

Each test image is a 2048×2048 grayscale Hα full-disk image, and a submission gives one run-length mask per filament. The metric is Panoptic Quality (PQ) [5]: a prediction and a ground-truth (GT) filament match when their IoU exceeds 0.5, and

$$\mathrm{PQ} = \frac{\sum_{\mathrm{TP}} \mathrm{IoU}}{\mathrm{TP} + \frac{1}{2}\mathrm{FP} + \frac{1}{2}\mathrm{FN}}.$$

Each annotator's labelling of an image (an *entry*) is scored separately against the same predictions, and the counts are pooled. Our main finding is that the labels limit the score: when we score each annotator against the other annotators of the same image, PQ is only 0.340 (SQ 0.634, RQ 0.536). Our design follows from this: learn the consensus, combine two resolutions, and reject candidates that annotators do not mark.

2 METHODOLOGY

2.1 Data, sampling and splits

The training set has 707 images and 8,199 filament polygons from 37 annotators; 411 images have one annotation set, 145 have two and 151 have three (1,154 entries). Filaments are small: the median polygon covers 1,228 pixels (0.03% of the image). We use five folds grouped by image, so all entries of an image stay in one fold. Filaments last several days, so we checked whether random folds are too easy: 28% of validation images have a training image within one day, but for the test images the share is 36%. Random folds therefore match the test situation, and folds by week would be harsher than the test. All numbers below are out of fold.

During training, 70% of crops are centred on a random filament pixel and 30% are placed uniformly. Augmentations are flips, 90-degree rotations and small brightness and contrast changes. We ported the organisers' PQ to run-length masks; a unit test checks that it matches their notebook to 10^{-6}.

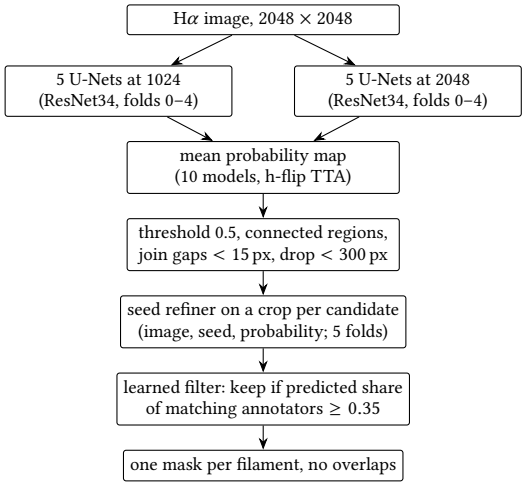


Figure 1: The end-to-end inference pipeline. Only the pixels of the test image are used.

2.2 Pipeline

Fig. 1 shows the pipeline. It uses existing building blocks: U-Net [6] with a ResNet34 encoder [3] (ImageNet start, segmentation-models-pytorch [4]). The seed refiner is our own design.

First stage. Each U-Net has one input channel. Its target is the fraction of annotators that marked each pixel, a soft label in $[0, 1]$; the loss is binary cross-entropy plus soft Dice. Per fold we train a *1024 model* (image downscaled to 1024, random 512^2 crops, batch 8, 60 epochs) and a *2048 model* (full resolution, 1024^2 crops, batch 4, 60 epochs). Both crops cover the same part of the disk; the 2048 model sees four times the detail. At inference we average the logits

[1] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

of the image and its horizontal flip, then the probability maps of all ten models.

Candidates (post-processing). The mean map is thresholded at 0.5. Connected regions closer than 15 pixels form one candidate, which joins fragments without changing any mask pixel; candidates under 300 pixels are removed. These three values were chosen on folds 0 and 1 from 108 settings and then fixed.

Seed refiner. Most first-stage errors are candidates on unmarked places or with too little overlap, not merges (Section 3.3). The refiner looks at one candidate at a time: a square crop (1.5 times its bounding box, at least 128 pixels, resized to 256) with three channels, namely the image, the candidate mask (the *seed*) and the first-stage probability. Its target is, for each annotator, the union of the GT filaments that touch the candidate, averaged over the annotators; a candidate that touches no GT filament has an empty target, so the refiner learns to reject it. The refiner is a three-channel ResNet34 U-Net, trained for 30 epochs on the candidates of the other four folds, which come from out-of-fold maps and so look like test candidates. At inference the five fold refiners are averaged; each contested pixel goes to the candidate with the higher probability, so masks never overlap.

Learned candidate filter. The refined pipeline still has about five false positives per image, many on places that no annotator marked. A LightGBM regressor predicts, for each refined candidate, the share of the image's annotators whose filament it matches at IoU above 0.5, which is what PQ counts. Its 17 features describe shape (area, bounding box, elongation, solidity), first-stage probability (mean, maximum, 90th percentile, mean in a 20-pixel ring), darkness (inside, ring, contrast and spread, relative to the disk median), the radial position on the disk and the number of candidates in the image. It learns from the 5,081 out-of-fold refined candidates. The features use the mean map of one fold's 1024 and 2048 models, as in training; at test time each candidate is scored with each fold's map and the five scores are averaged. Candidates with a score below 0.35 are removed.

3 EVALUATION

3.1 Strategy

We judge every change by out-of-fold PQ with the fixed post-processing. Two training seeds of the same model differ by up to 0.004 on one fold, so we kept a change only if it gained more than 0.007 on two folds. The filter is judged by nested cross-validation: for each outer fold, it is trained on the other four folds, its threshold is chosen by an inner 4-fold cross-validation on those folds, and both are applied to the outer fold, so no score uses a threshold tuned on its own images. For the final pipeline we report per-fold scores, the spread over folds, and a bootstrap over images (2,000 samples) for pooled PQ and for the paired gains. The organisers' own plotting code draws the distributions and matching counts.

3.2 Results

Table 1 and Fig. 3 give the scores. The fold spread (0.007–0.010) is larger than the gains, because folds differ in difficulty; the paired comparisons are clearer. The 1024 + 2048 mean beats the 1024 model on every fold (+0.013 on average). The refiner gains on four folds and is neutral on fold 2; the filter gains on all five folds (+0.006 to +0.018). Pooled over all 707 images, PQ is 0.426 [0.416, 0.435] for the first stage, 0.431 [0.422, 0.440] after the refiner and 0.445 [0.435, 0.454] after the filter (95% bootstrap intervals). The paired gains are +0.0054 [0.0032, 0.0078] for the refiner and +0.0136 [0.0106, 0.0168] for the filter, both positive in all 2,000 samples. These intervals cover the choice of images, not training noise (about 0.003 between refiner seeds). On the public leaderboard the four pipelines scored 0.35, 0.36, 0.37 and 0.38; the public score never moved against the local score and stays about 0.06 below it. On the 296 images with two or three annotators, our masks score PQ 0.446 (SQ 0.677) against each annotator, while one annotator scores 0.340 (SQ 0.634) against the others: the model agrees with each annotator more than the annotators agree with each other.

Table 1: Out-of-fold PQ per fold, mean and standard deviation over folds.

Pipeline	F0	F1	F2	F3	F4	mean ± sd
1024 model	.410	.421	.406	.406	.422	.413 ± .008
2048 model	.415	.431	.422	.423	.431	.425 ± .007
1024 + 2048	.420	.434	.419	.423	.436	.426 ± .008
+ seed refiner	.427	.440	.419	.428	.443	.432 ± .010
+ learned filter	.446	.446	.435	.442	.456	.445 ± .007

Fig. 4 compares the first stage with the final pipeline. Refiner and filter together remove 1,411 false positives (3,437 → 2,026) at the cost of 350 more false negatives (2,870 → 3,220). The GT filaments that overlap two or more predictions fall from 394 to 242; the predictions that overlap two or more GT filaments stay about the same (257 → 262; the refiner alone raised them to 281). The mean IoU of overlapping pairs rises from 0.570 to 0.590 (Dice 0.703 → 0.722).

3.3 Strengths, weaknesses and possible improvements

Strengths. The pipeline is simple, has few tuned values, and every choice was validated out of fold on two or more folds. It runs at 1.5 s per image on one RTX 3070 (0.83 s for the 2048 models) and 79 s on four CPU threads. On a GPU it is deterministic: the notebook in our repository rebuilds the submitted CSV with the same SHA-256 from the stored weights on the same GPU type; on another GPU type the masks agree with a mean IoU of 0.9998, and on a CPU with 0.997.

Weaknesses: where the remaining PQ goes. Table 2 splits the errors of the final pipeline (out of fold, all 707 images; `scripts/gap_analysis`). Of the 1,878 GT filaments that no prediction touches, 683 have no first-stage pixel above 0.5, 574 have fewer than 300 such pixels, and 271 were found by the refiner but removed by the filter, which pays for 1,000 fewer false positives. On images with two or three annotators, only 56% of the missed filaments are also marked by another annotator, against 93% of the matched ones: about half of the misses are one annotator's choice (Fig. 5, middle). Near misses go both ways: the prediction is smaller than the GT in 462 cases and larger in 370; 129 GT filaments are split and 189 predictions span two GT filaments (Fig. 5, right). Growing or shrinking every

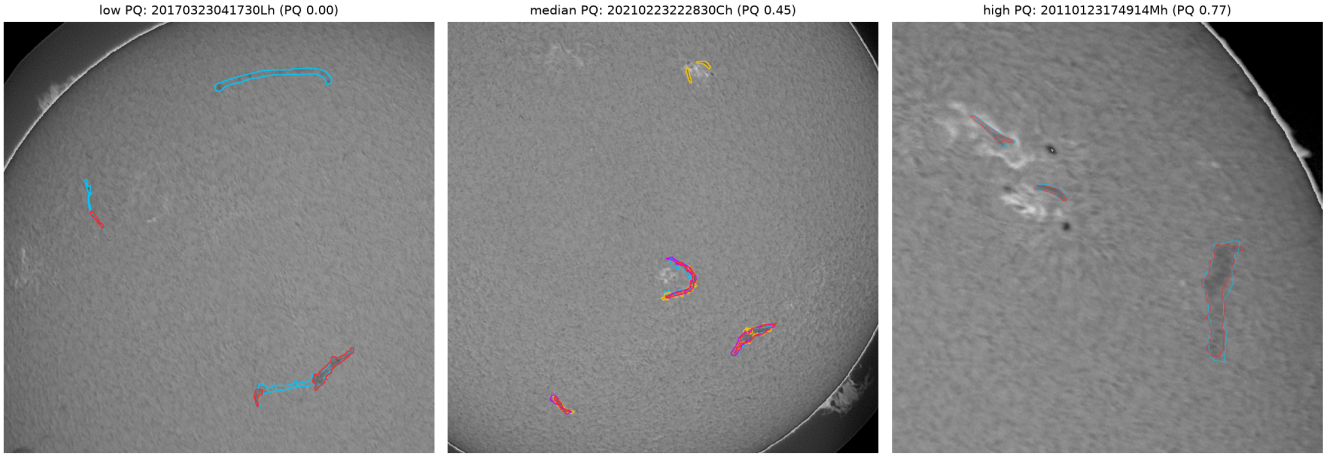


Figure 2: Out-of-fold example segmentations with low, median and high per-image PQ. Annotators in cyan, yellow and magenta; our masks in red. Left: the annotator marks a long faint filament that we miss, and our masks cover other parts of the marked filaments than the annotator's outlines. Middle: three annotators and our masks agree on three filaments; one annotator also marks a small filament at the top that the others and our model leave out. Right: close agreement.

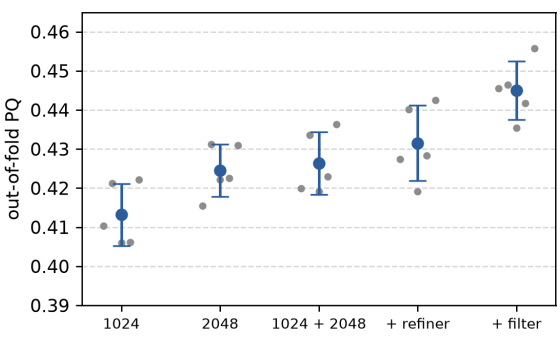


Figure 3: Out-of-fold PQ per pipeline: mean ± standard deviation over the five folds (blue) and each fold (gray).

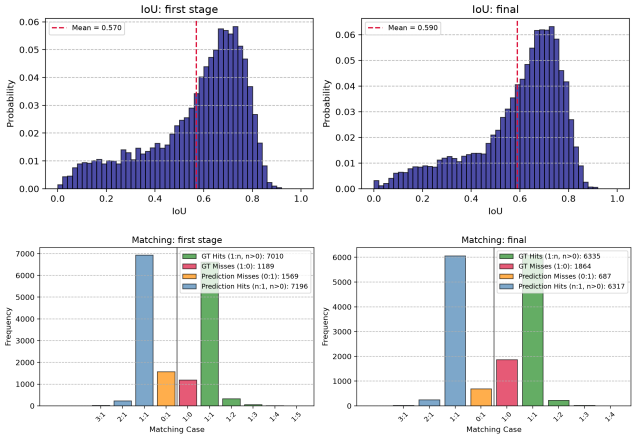


Figure 4: Out of fold, all 707 images, drawn with the organisers' plotting code. Top: IoU of overlapping (GT, prediction) pairs. Bottom: matching cases; "2:1" is a prediction that overlaps two GT filaments, "1:2" a GT filament that overlaps two predictions.

Table 2: Error types of the final pipeline (out of fold) and the PQ if every case of the type were fixed (missed GT found at IoU 0.6, near misses at IoU 0.55, false positives removed).

Error type	count	PQ if fixed
Final pipeline (TP)	4,979	.445
GT never touched (IoU < 0.05)	1,878	.528
Near-miss pairs (IoU 0.3–0.5)	916	.510
GT with weak overlap (0.05–0.3)	426	–
FP with no GT nearby	697	.466

mask by one pixel lowers PQ to 0.418 and 0.417, so there is no global bias to correct.

What did not help (all out of fold, same post-processing): a majority-vote or random-annotator target instead of the soft mean (−0.003, −0.005); a mean-probability filter on candidates (+0.003); eight flips and rotations as test-time augmentation (+0.004 and −0.003 on two folds, at four times the cost); a ResNet50 encoder at 1024 (+0.004 as a third model, +0.006 instead of the ResNet34); averaging two refiner seeds (no gain); a focal + Dice loss (−0.005); six more filter features (two per-resolution probabilities, their difference, edge strength, ring spread, dark quantile: −0.0004); an auxiliary spine output trained on the MAGFiLO spines (+0.005 and +0.003 for the 1024 model alone, but only +0.001 inside the 1024 + 2048 mean); more candidates for the filter (first-stage threshold 0.4, minimum area 150: ±0.000 pooled); input cleaning as in EdgeAttNet [7] (disk mask, radial flattening, CLAHE: −0.004 and −0.002); an auxiliary

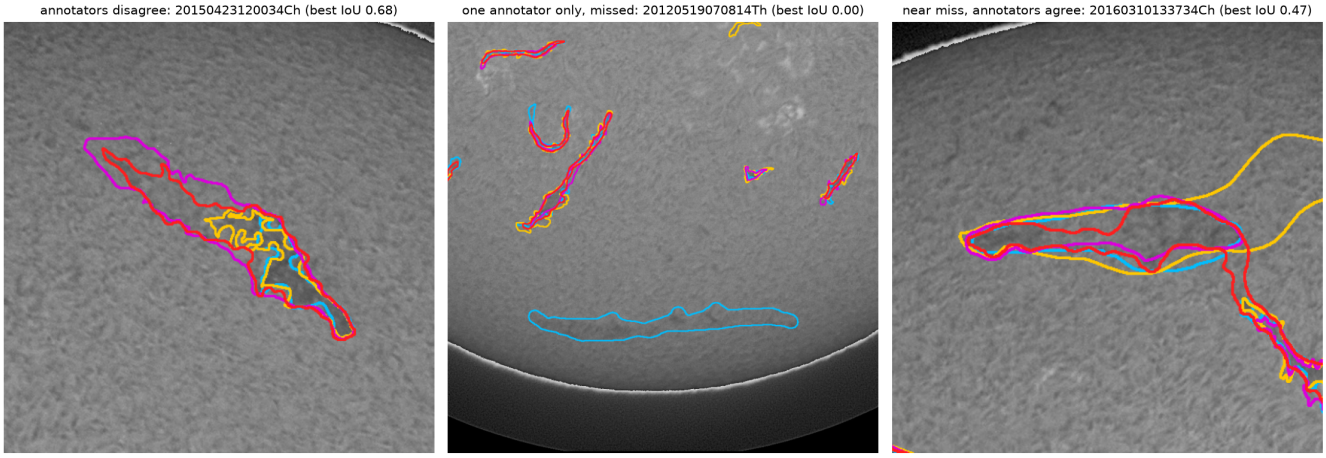


Figure 5: Zoomed out-of-fold cases from images with three annotators (cyan, yellow, magenta; our masks in red), each the largest GT filament of its kind. Left: the annotators' outlines of one filament differ widely (best agreement IoU 0.31) and our mask lies between them. Middle: one annotator marks a long faint filament near the limb that the other two and our model leave out. Right: a near miss where the annotators agree (IoU 0.72): they split the filament from its lower arm, our mask joins them (IoU 0.47).

output for the contact zone between filaments (the zone is present in only 121 of 707 images and the output stays at zero); and other weights than 0.5 for the 2048 map (best 0.5).

Possible improvements. A refiner that uses the spine to cover the whole filament (spine supervision of the first stage did not help); an instance signal that reaches further than a contact zone, such as a per-filament embedding, to split masks that span two filaments; and self-supervised pretraining on unlabeled Hα archives, which the organisers allow when it excludes the test observations.

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